

AD 502 – Machine Learning Lab

Pre-Reading Material – Lecture 3

K-Nearest Neighbour

Estimated Reading Time: 25–30 minutes

Introduction

K-Nearest Neighbors (KNN) is a simple and popular **supervised machine learning algorithm** used mainly for **classification** and also for regression.

The basic idea is:

Similar data points are likely to have similar outcomes.

For example, if we want to predict whether a person will purchase a product, KNN looks at people with similar characteristics, such as **age and estimated salary**, and uses their outcomes to make a prediction.

How KNN Works

KNN follows a simple process:

1. A new data point is given to the model.
2. KNN looks for the **K closest data points** in the training data.
3. These are called the **nearest neighbors**.
4. For classification, the most common class among these neighbors is selected as the prediction.

Simple Example

Suppose we want to classify a new customer as:

- **Purchased**
- **Not Purchased**



If $K = 5$, KNN looks at the 5 customers most similar to the new customer.

If 3 of them purchased the product and 2 did not, the prediction will be:

Purchased

What Does K Mean?

K represents the number of neighbors considered when making a prediction.

For example:

- $K = 1 \rightarrow$ considers only the closest data point
- $K = 3 \rightarrow$ considers the 3 closest points
- $K = 5 \rightarrow$ considers the 5 closest points

The choice of K can affect the model's performance.

- Very small $K \rightarrow$ model may be sensitive to individual data points.
- Very large $K \rightarrow$ model may become less sensitive to local patterns.

Therefore, an appropriate value of K is usually selected by testing different values.

Importance of Feature Scaling

KNN determines which points are "close" to each other.

Therefore, **feature scaling is very important**.

Consider two features:

- Age: 20–60
- Salary: 20,000–1,00,000

Salary has much larger numerical values than age. Without scaling, salary may have a much greater influence when determining distance.

A common approach is **StandardScaler**, which puts features on a comparable scale.

KNN in Classification

KNN can be used for problems such as:

- Customer purchase prediction
- Disease classification
- Spam detection
- Image classification
- Recommendation-related tasks

For example, in a customer dataset, features such as **Age** and **Estimated Salary** can be used to predict whether a customer will **Purchase** a product.

Training and Testing

The dataset is generally divided into:

- **Training data** – used to build the model
- **Testing data** – used to check how well the model performs on unseen data

The test data should not be used for training the model.

This helps us understand whether the model can make predictions on new data.

Evaluating KNN

After making predictions, we can evaluate the model using measures such as:

Accuracy

Shows the proportion of predictions that are correct.

Confusion Matrix

Shows how many predictions were:

- Correctly classified
- Incorrectly classified

Classification Report

Provides commonly used measures such as:

- Precision
- Recall
- F1-score

These measures help us understand the classification performance more clearly.

Key Points to Remember

- KNN is a **supervised learning algorithm**.
- It can be used for **classification and regression**.
- It makes predictions using the **nearest data points**.
- **K** represents the number of neighbors considered.
- Feature scaling is important for KNN.
- A suitable value of K should be selected carefully.
- Performance can be evaluated using **accuracy, confusion matrix, precision, recall, and F1-score**.

Quick Pre-Reading Questions

1. What type of machine learning algorithm is KNN?
2. What does **K** represent?
3. Why is feature scaling important in KNN?
4. What happens when $K = 1$?
5. What is the purpose of a confusion matrix?
6. Give one real-world application of KNN.

